Chapter 1. Introduction to Nonparametric Statistics

Nonparametric models, parametric inference review, why nonparametric methods, robustness

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1.1 Parametric and nonparametric models

A statistical model is a family $\mathcal{F}$ of distributions assumed to contain the population. It is **parametric** if it is indexed by finitely many real numbers, e.g. the normal family

$$\mathcal{F}_G = \{N(\mu, \sigma^2) : \mu \in \mathbb{R},\ \sigma > 0\}. \tag{1.1}$$

It is **nonparametric** if no finite-dimensional parametrization exists. Typical examples:

$$\mathcal{F}_1 = \{F : \int x^2\, dF < \infty\}, \qquad \mathcal{F}_{\text{all}} = \{\text{all } F\},$$

$$\mathcal{F}_2 = \{F : F \text{ has a density symmetric about some } \theta\}.$$

Both $\mathcal{F}_G \subset \mathcal{F}_1$ and $\mathcal{F}_G \subset \mathcal{F}_2$. A model with a finite-dimensional part of interest and an infinite-dimensional nuisance part, such as $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{i.i.d.}}{\sim} F$ unrestricted, is called **semi-parametric**.

In a nonparametric model a **parameter** is a functional $\theta = \theta(F)$: the mean $\int x\, dF$, the median, a quantile, the density f, or a regression curve $m(x) = E(Y \mid X = x)$. A procedure whose validity (e.g. type I error) holds for every F in a very large family is **distribution-free**.

Three tools:

1. **Ranks, signs and permutation tests** (Part I): conditional inference given the observed values, exact under exchangeability.
2. **The bootstrap** (Part II): replace the unknown F by an estimate and compute sampling distributions by simulation.
3. **Smoothing** (Part II): estimate a function $f(x)$ or $m(x)$ by averaging the observations near x.

1.2 Review of parametric inference

Motivation. We recall the t-test and likelihood framework to see which assumptions each rank method drops.

Definition (Quantile)

For a distribution function F and $p \in (0,1)$, the p-quantile is $Q(p) = \min\{x : F(x) \geq p\}$. For the standard normal, $z_\alpha := \Phi^{-1}(1-\alpha)$.

Proposition 1.1

Let F be the distribution function of an integer-valued X (e.g. $X \sim B(n,p)$). Then $q = Q(p)$ is the smallest integer with $F(q) \geq p$, and $F(q-1) < p \leq F(q)$. In particular $P(X \leq q) \geq p$ and $P(X \geq q) > 1 - p$.

Proof. F is nondecreasing, right-continuous and constant on each $[k, k+1)$, so $\{x : F(x) \geq p\} = [q, \infty)$ with q the smallest integer satisfying $F(q) \geq p$. Minimality gives $F(q-1) < p$. Then $P(X \geq q) = 1 - F(q-1) > 1 - p$. ■

If $X \sim N(\mu, \sigma^2)$, $F_X(x) = \Phi((x-\mu)/\sigma)$. If $X \sim B(n,p)$, $EX = np$, $\operatorname{Var} X = np(1-p)$, and for $np \geq 10$ roughly $X \approx N(np, np(1-p))$ by the CLT.

Estimation. An estimator $\hat{\theta}$ is judged by its bias $E\hat{\theta} - \theta$, standard error $\mathrm{se}(\hat{\theta}) = \sqrt{\mathrm{Var}\,\hat{\theta}}$ and mean squared error

$$\mathrm{MSE}(\hat{\theta}) = E(\hat{\theta} - \theta)^2 = \mathrm{bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}).$$

Under $\mathcal{F}_G$, $\bar{X}_n$ is the best unbiased estimator of μ, and $T = \sqrt{n}(\bar{X}_n - \mu)/s \sim t_{n-1}$ gives the exact interval $\bar{X}_n \pm t_{\alpha/2,n-1}s/\sqrt{n}$; outside $\mathcal{F}_G$ both properties are lost.

Testing. A test rejects when a statistic T falls in a region with $P_{H_0}(\text{reject}) \le \alpha$ (**level**); we want large **power** $P_{H_1}(\text{reject})$. The p**-value** is the H_0-probability of a value of T at least as extreme as observed; $p \le \alpha$ iff T lies in the level-α region. Nonparametric problems need a statistic whose **null distribution** does not depend on F.

Proposition 1.2 (Duality of tests and intervals)

Suppose for each θ_0 there is a level-α test ϕ_{θ_0} of $H_0 : \theta = \theta_0$. Then $C = \{\theta_0 : \phi_{\theta_0} \text{ does not reject}\}$ is a $1-\alpha$ confidence set. Conversely, if C is a $1-\alpha$ confidence set, rejecting $H_0 : \theta = \theta_0$ when $\theta_0 \notin C$ is a level-α test.

Proof. If θ is the true value, $P(\theta \in C) = P(\phi_\theta \text{ does not reject}) \ge 1-\alpha$. Conversely $P_{\theta_0}(\theta_0 \notin C) \le \alpha$. ■

The central limit theorem.

Theorem 1.3 (CLT)

If $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ with mean μ and variance $0 < \sigma^2 < \infty$, then $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$.

Proof [Supplement]. Let $Y_i = (X_i - \mu)/\sigma$ and $\varphi(t) = Ee^{itY_1}$. Since $EY_1^2 < \infty$, $\varphi(t) = 1 - \frac{t^2}{2} + o(t^2)$. By independence $Ee^{itZ_n} = \varphi(t/\sqrt{n})^n = (1 - \frac{t^2}{2n} + o(\frac{1}{n}))^n \to e^{-t^2/2}$, the characteristic function of $N(0,1)$; Lévy's continuity theorem concludes. ■

So t-procedures need only the *statistic* to be approximately normal. But the CLT gives no rate: convergence is slow for skewed F and fails when $\sigma^2 = \infty$ (Cauchy, slash), the first motivation for nonparametric methods.

1.3 Why nonparametric methods

- *Validity under weak assumptions.* Rank, sign and permutation tests control the type I error exactly for every F in a large family, at any sample size.
- *Computation instead of derivation.* Permutation and bootstrap distributions replace formulas that are unknown or valid only asymptotically.
- *Flexibility.* Function-valued targets (f, m) are estimated without assuming a shape.
- *Robustness.* Procedures based on ranks, signs or medians are insensitive to outliers.

Example 1.1 (Effect of one outlier)

Take 2.1, 2.4, 2.6, 2.9, 3.0, 3.2, 3.5, 25. With/without the last value: mean 5.59 / 2.81, standard deviation 7.86 / 0.48, median 2.95 / 2.9, and **median absolute deviation** MAD $= \mathrm{median}_i|x_i - \mathrm{median}_j\, x_j|$ 0.45 / 0.30. One point dominates the mean and SD; the median and MAD barely move.

Distribution-free methods keep their properties under heavy tails (where outliers come from), so they are typically robust; the converse need not hold.

1.4 Selected exercises

Motivation. Exercises on quantiles and CLT-based inference, and on exact pivots that replace the normal approximation.

Exercise 1.1 (Normal quantiles)

Express the quantile function $Q(p)$ of $N(\mu, \sigma^2)$ through Φ, and find the median $Q(0.5)$ and the quartiles $Q(0.25)$, $Q(0.75)$.

Solution.

Step 1. $F(x) = \Phi((x-\mu)/\sigma)$ is continuous and strictly increasing on $\mathbb{R}$ with limits 0 and 1. So for $p \in (0,1)$ there is a unique x_p with $F(x_p) = p$, and $\{x : F(x) \geq p\} = [x_p, \infty)$. Hence $Q(p) = x_p$.

Step 2. $F(x_p) = p$ means $(x_p - \mu)/\sigma = \Phi^{-1}(p)$, so

$$Q(p) = \mu + \sigma\, \Phi^{-1}(p).$$

Step 3. Since ϕ is even, $\Phi(-z) = 1 - \Phi(z)$, so $\Phi^{-1}(1-p) = -\Phi^{-1}(p)$ and $\Phi^{-1}(0.5) = 0$. With $\Phi^{-1}(0.75) = 0.6745$,

$$Q(0.5) = \mu, \qquad Q(0.25) = \mu - 0.6745\,\sigma, \qquad Q(0.75) = \mu + 0.6745\,\sigma.$$

So the normal IQR is $1.349\,\sigma$: the source of the divisor 1.34 in Section 11.2 and of the MAD factor $1/0.6745 = 1.4826$ (Section 10.3.1). ■

Exercise 1.5 (Exponential waiting times)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(\theta)$ with mean $\theta > 0$, i.e. density $\theta^{-1} e^{-x/\theta}$ for $x > 0$.
(a) Use the CLT to find the approximate distribution of $\bar{X}_n$.
(b) The waiting time for a bus is modeled this way, and $n = 40$ students waited $\bar{x} = 7$ minutes on average. Give an approximate 95% confidence interval for θ, and test $H_0 : \theta = 10$ against $H_1 : \theta < 10$; report the p-value and the conclusion at $\alpha = 0.01$.

Solution.

(a) With $y = x/\theta$, $EX = \theta \int_0^\infty y e^{-y}\, dy = \theta$ and $EX^2 = \theta^2 \int_0^\infty y^2 e^{-y}\, dy = 2\theta^2$, so $\text{Var}\, X = \theta^2$. By Theorem 1.3, $\sqrt{n}(\bar{X}_n - \theta)/\theta \xrightarrow{d} N(0,1)$, i.e. $\bar{X}_n$ is approximately $N(\theta, \theta^2/n)$.

(b) *Interval.* By the LLN $\bar{X}_n/\theta \xrightarrow{p} 1$, so by Slutsky $\sqrt{n}(\bar{X}_n - \theta)/\bar{X}_n \xrightarrow{d} N(0,1)$, giving

$$\bar{x} \pm 1.96\, \frac{\bar{x}}{\sqrt{n}} = 7 \pm 1.96 \cdot \frac{7}{\sqrt{40}} = 7 \pm 2.17 = (4.83,\ 9.17).$$

Test. Under H_0 the SD $\theta_0/\sqrt{n}$ is known, so use $Z = \sqrt{n}(\bar{X}_n - \theta_0)/\theta_0$, rejecting for small Z: $z = \sqrt{40}(7-10)/10 = -1.90$, $p = \Phi(-1.90) = 0.029 > 0.01$. Not rejected at 1% (rejected at 5%). ■

Exercise 1.7 (Exact tests under a model)

Can the test of Exercise 1.5 be carried out exactly, without the CLT, if the population is assumed to be (a) Exp(θ), or (b) Uniform$(0, 2\theta)$ (which also has mean θ)?

Solution.

(a)

Step 1. (A pivot.) $Y_i = 2X_i/\theta$ has the χ^2_2 density $\frac{1}{2}e^{-y/2}$, with m.g.f. $(1-2t)^{-1}$, $t < \frac{1}{2}$; so $\sum_i Y_i$ has m.g.f. $(1-2t)^{-n}$, that of χ^2_{2n}:

$$V(\theta) = \frac{2n\bar{X}_n}{\theta} \sim \chi^2_{2n} \qquad \text{for every } \theta > 0.$$

Step 2. (Test.) Under H_0, $V(\theta_0) \sim \chi^2_{2n}$; for $\theta < \theta_0$, $V(\theta_0) = (\theta/\theta_0)V(\theta)$ is stochastically smaller, so reject for small $V(\theta_0)$: $p = P(\chi^2_{2n} \le 2n\bar{x}/\theta_0)$. Here $v = 56$ and $p = P(\chi^2_{80} \le 56) = 0.019$, not significant at 1%.

Step 3. (Interval.) With q_p the p-quantile of χ^2_{2n}, $P(q_{0.025} \le V(\theta) \le q_{0.975}) = 0.95$ gives the exact interval $\left(2n\bar{X}_n/q_{0.975},\ 2n\bar{X}_n/q_{0.025}\right) = (5.25,\ 9.80)$. It lies right of the CLT interval and the p-value is smaller, reflecting the right skew of $\bar{X}_n$.

(b)

Step 1. (A pivot.) Let $M = \max_i X_i$. For $0 \le m \le 2\theta$, $P(M \le m) = (m/2\theta)^n$, so $U(\theta) = (M/2\theta)^n$ has $P(U(\theta) \le u) = P(M \le 2\theta u^{1/n}) = u$: $U(\theta) \sim$ Uniform$(0, 1)$.

Step 2. (Test.) Under $\theta < \theta_0$ all observations lie below 2θ, so small M supports H_1: $p = P_{\theta_0}(M \le m_{\text{obs}}) = \min\{1, (m_{\text{obs}}/2\theta_0)^n\}$. Under H_0, $p = U(\theta_0)$ is uniform, so $P(p \le \alpha) = \alpha$: exact.

Both exact tests use a model-specific statistic ($\bar{X}_n$ with the χ^2 law, or M; $\bar{x}$ alone does not suffice for (b)) and are exact only under their own model, a dependence nonparametric methods avoid. ■